Dr Jennifer L. Hawkin

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Personal Profile

Interdisciplinary researcher and engineer specialising in resource efficiency, industrial decarbonisation, and sustainable materials and infrastructure systems. Combines systems modelling and environmental assessment with eight years of industrial engineering experience at Rolls-Royce. Current research examines the feasibility, resource requirements, and environmental implications of net-zero and circular-economy pathways across material-intensive sectors.

Research Interests: industrial ecology - resource constraints - energy systems - circular economy - plastics and chemicals - climate mitigation - systems modelling

Career and Education Overview

2024-present **Postdoctoral Researcher**, *University of Sheffield, UK (School of Chemical, Materials & Biological Engineering)*, Resource efficiency and environmental impacts of petrochemical consumption, material production, and energy infrastructure

- Lead and contribute to peer-reviewed publications on polymer end-of-use, carbon circularity, and science-based targets for materials.
- Coordinated the management and reporting of a Royal Society-funded collaboration with colleagues at the National Institute for Environmental Studies, Japan.
- Presented findings to industry contacts and academic audiences.

2021-2024 **PhD**, *University of Cambridge, UK (Engineering Dept.)*, **Supervisor:** Professor Julian Allwood, **Title:** Do Net-Zero plans add up? A framework and model to quantify risks of resource supply shortages in climate mitigation strategies

2012-2021 **Industrial Experience**, *Rolls-Royce & Engineers Without Borders*
UK, Singapore, Malaysian Borneo (See Industry Experience below.)

2008-2012 **BA & MEng (Hons) Engineering**, *University of Cambridge, UK*, **Honours with Distinction**, specialised in Mechanical Engineering; included one year studying at Massachusetts Institute of Technology (M.I.T.), USA

Selected undergraduate awards: Lancaster Prize for outstanding performance in Engineering, Foundation Scholarship, and Foundress Prize (Pembroke College); Technology Partnership Prize for Integrated Design (Department of Engineering).

Industry Experience

2017-2021 **Rolls-Royce plc, UK**, *Roles included: Technical Assistant to the Chief Technology Officer, Sub-System Integration & Component Design, Innovation Technology Lead.*

- Supported the Chief Technology Officer with technology strategy, executive briefings, and cross-business technical coordination.
- Worked across subsystem integration, component design, and innovation to evaluate and progress emerging technologies.

2016-2017 **TONIBUNG (Friends of Village Development), Malaysian Borneo**,
EWB-UK (Engineers-Without-Borders UK) Fellowship Volunteer

- Collaborated with local and international colleagues to develop and roll out novel technology for clean electricity generation in remote communities

2016 **Rolls-Royce, Singapore**, *Regional Coordinator (Secondment)*
- Scoped and drafted the proposal for a government-funded research centre in Singapore

2012-2016 **Rolls-Royce plc, Derby/Bristol/Singapore**,
Various roles in Supply Chain and the Strategic Research Centre
- Projects included computational and experimental fluid flow characterisation and cross-sector work, taking a strategic view of emerging technologies.

Teaching Experience

2026 **Guest Lecturer**, *University of Sheffield (Engineering)*, Undergraduate level
- Delivered an interactive lecture, titled "The use and misuse of environmental metrics", introducing students to environmental metrics and encouraging students to think critically about how tools are applied and results interpreted.

2025-2026 **Project Supervisor**, *University of Sheffield*, Master's-level student projects
- Comparative Life Cycle Assessment of Elium and Conventional Epoxy Systems for Turbine Blade Manufacturing
- Assessing the future increased polymer demand in the transition to domestic Battery Electric Vehicles within the UK

2025 **Tutor**, *University of Cambridge (Engineering)*, Master's-level course
- "Climate Change Mitigation". Provided guidance on, and finally marked poster submissions of, student proposals for climate mitigation options. Student projects must consider technical potential of proposals and rates of deployment within the whole system of global emissions. Student projects span all economic and industrial sectors.

2025 **Teaching Assistant**, *Beijing Normal Summer School*, Postgraduate level:
- Provided student support to understand and progress through exercises, including techno-economic analysis (TEA), life cycle assessment (LCA) and material l=flow analysis (MFA).

2021-2024 **Tutor and Facilitator**, *University of Cambridge (Engineering)*, Masters and undergraduate-level courses
- "Structural Design". Supported students to understand engineering concepts behind simple truss structures, and in the design process to design, build and test a metal cantilever or bridge. Marked and provided feedback for all student reports.
- "Sustainable Engineering". Facilitated discussions on sustainable engineering. Student topics could consider any aspect of sustainability and often spanned across economic and industrial sectors.
- "Climate Change Mitigation" (as above)

2022-2023 **Supervisor**, *St Catharine's College, University of Cambridge*, Second-year undergraduate course: Materials thermodynamics & diffusion and materials processing
- Led small-group tutorials for second-year undergraduates, guiding students through course concepts and problem-solving.

2022-2023 **Interviewer**, *St Catharine's College, University of Cambridge*, Undergraduate admissions (Engineering)

Selected Conference and Event Participation

2026 **Oral & Poster Presentations**
- "Plastic Pathways for the UK: Mapping plastics production, trade, consumption and waste flows" and "Rethinking circularity in petrochemicals: why targeting carbon loops is not enough", presented at *the Industrial Society for Industrial Ecology (ISIE) Socio-Economic Metabolisms (SEM) Conference (Cambridge UK)*

2025 **Oral & Poster Presentations**
- "Re-evaluating plastics end-of-use treatment options" and "Engineering Risk Assessments for Climate Policy - Can Failure Modes and Effects Analysis (FMEA) improve the prospects of Carbon Dioxide Removal (CDR)?", presented at *the Industrial Society for Industrial Ecology (ISIE) Conference (Singapore)*
- "Do Net-Zero plans add up?", presented at *the World Resources Forum (online)*
- "Constraints on carbon circularity in the petrochemical sector", presented at *the Annual EDITS Meeting (Energy Demand changes Induced by Technological and Social innovations), IIASA, Laxenburg, Austria*

2024 **Oral Presentation**, *Annual EDITS Meeting (Energy Demand changes Induced by Technological and Social innovations)*, IIASA, Laxenburg, Austria
"A model and interactive calculator to aggregate demands of climate mitigation proposals"

Invited participant, *UNEP & Life Cycle Initiative Biogenic Carbon in LCA Guidance Project Workshop*, Bath, UK

Poster, *ISIE (Industrial Society for Industrial Ecology) Gordon Research Conference*, Les Diablerets, Switzerland
"Is Net-Zero at Risk? Modelling aggregated energy demands of climate mitigation plans reveals high risk 2050 expectations."

Invited discussion leader, *ISIE Gordon Research Seminar*, Les Diablerets
"Decarbonization Across Sectors"

2023 **Invited speaker**, *Panel discussion: Clearing the air - information literacy in the climate crisis*, Cambridge University

Oral Presentation, *Symposium: "Governance, law and economics of climate change and energy transition"*, Cambridge University
"What scale of negative emissions can we rely on?"

Oral Presentation, *ISIE Conference*, Leiden, The Netherlands
"A framework for aggregating energy demands of net-zero proposals"

2022 **Oral Presentation**, *Cambridge Zero Research Symposium*
"Should 'we' change our behaviour to reach net-zero?"

2017 **Invited speaker**, *Institution of Mechanical Engineers, Singapore Network*
"Micro-Hydro Power - Clean and Affordable Electricity in the Jungle"

2015 **Invited speaker**, *Institution of Engineering and Technology East Midlands Network*
"Are Unmanned Autonomous Vessels the Future for Commercial Shipping?"

Professional Development and Institutional Engagement

2025 **Runner-up, Best Poster**, *ISIE Conference (Singapore)*

2022-2024 **Equity, Diversity & Inclusion (EDI) steering group**, *Darwin College, Cambridge*

2020-2022 **Ethical principles committee**, *Institution of Mechanical Engineers (IMechE)*

2018-2022 **Registered chartered engineer (CEng)**, *IMechE*

2016 **Volunteering and personal development award**, *IMechE*

Additional Skills

Methods and computational skills:
Systems modelling; life-cycle assessment; material flow analysis; Python for model development and visualisation; R Markdown for documentation.

Languages:
- Conversational Italian
- Conversational German
- Basic French

Other activities and interests:
Recreational sports (including swimming, cycling, rambling and rock climbing); European folk dance; Scouting

Publications

2026 **Demand-side innovation is the priority for decarbonizing materials**, *Nature Reviews Materials*, Allwood, J.M., Baker, F.F., ..., & Hawkin, J.L., DOI: 10.1038/s41578-026-00934-2
By estimating the physical resource requirements, public finance needs, and levels of societal participation implied by different UK emissions trajectories, this analysis shows that demand reduction must be a central priority for UK mitigation policy.

Aggregating demand for three fundamental resources to avoid burden-shifting in climate policy, *Environmental Science & Technology*, Hawkin, J.L., & Allwood, J.M., DOI: 10.1021/acs.est.5c12742
By aggregating demand for emissions-free electricity, biomass and carbon storage required by climate policy proposals across sectors, we show they imply implausible deployment rates, unless demand is reduced through participatory innovations.

Stylized facts for the circular economy - a knowledge synthesis, *preprint*, Pauliuk, S., Galychyn, O., Hawkin, J.L., *et al.*, ssrn.com, abstract id=7028698
We present an initial set of circular economy (CE) stylized facts: empirically grounded statements that synthesize existing knowledge. Developed through collaborative review and synthesis, they make the CE literature accessible to diverse audiences while highlighting current strengths and gaps in research and implementation.

Reducing greenhouse gas emissions from polymer processing at end-of-use, *Journal of Cleaner Production*, Gao, Y., Meng, F., Hawkin, J.L., Cullen, J., & Cabrera Serrenho, A., DOI: 10.1016/j.jclepro.2026.148308
We quantify greenhouse gas emissions associated with end-of-use options for polymers, globally. Our findings reveal that enhancing recycling and landfilling could substantially reduce emissions by 30%, while mismanagement practices such as open burning and dumping exacerbate environmental pollution and must be avoided.

Rethinking residual plastic waste management in an emissions-constrained future), *Nature Reviews Materials*, Hawkin, J.L., Ryan, A., Cullen, J., & Meng, F.
We compare plastic waste treatment options, including mechanical recycling, advanced recycling, waste-to-energy technologies, conventional landfill, and a hypothetical "curated storage". We find that treatment pathways require more nuanced evaluation, including consideration of feedstock composition, process and infrastructure maturity, energy system context, governance constraints, and long-term liability.

Rethinking circularity in chemicals: why targeting carbon loops is not enough (under review), *Chem Circularity*, Hawkin, J.L., & Meng, F.
This paper responds to rapidly expanding net-zero and circular economy strategies that increasingly prioritise "carbon circularity" within the chemicals sector, such as recycling, alternative feedstocks, and carbon capture and utilisation. Rather than assuming that closing carbon loops will deliver transformative mitigation at scale, we synthesise evidence across deployment feasibility, energy and resource constraints, and circularity hierarchies to evaluate what these strategies can and cannot deliver in practice.

Science-based targets to balance environmental impacts and inequalities of material production and use (in preparation), Hawkin, J.L., Watari, T., Ryan, A., & Meng, F.
Drawing on existing scientific frontiers, preliminary science-based targets are proposed for four key materials: steel, cement, plastics, and sand.

2025 **End of life management for wind turbines**, *Nature Reviews Clean Technology*, Meng, F., Hawkin, J.L., Gast, L. *et al.*, DOI: 10.1038/s44359-025-00097-3
This Review assesses current and emerging EOL practices, comparing the environmental and economic trade-offs across mechanical, thermal and chemical recycling technologies to recover materials from wind turbines.

Treatment of temporary GHG removals in voluntary carbon markets, *Cambridge Open Engage*, Zschietzschmann, H., Hawkin, J.L., Rosser, J.P. *et al.*, DOI: 10.33774/coe-2025-5sbl5
This report provides a broad introduction to the issue of carbon storage permanence, equivalence and accounting approaches.

2024 **Do net-zero plans add up? A framework and model to quantify risks of resource supply shortages in climate mitigation strategies**, *PhD Thesis*, Hawkin, J.L., DOI: 10.17863/CAM.119203
This thesis develops a framework to aggregate resource demands of net-zero proposals by modelling low-carbon and conventional processes across industry and domestic sectors. Application of the model shows that dominant mitigation strategies rely on implausible deployment rates of non-emitting electricity generation and carbon storage and/or unsustainable use of biomass.